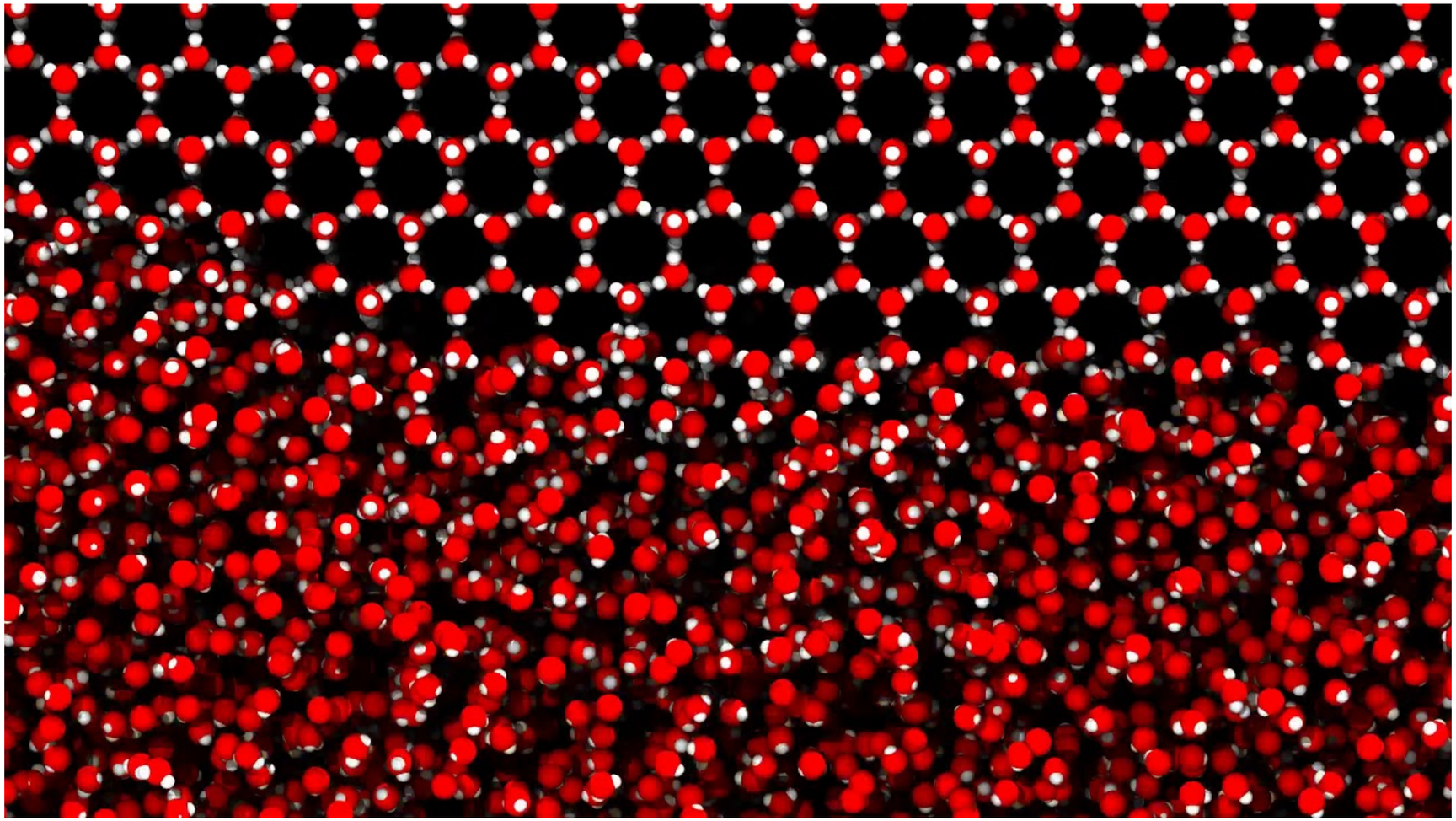
Computational Methods in Material Science

MS2860/MS5860 (S-slot, 36 Segment)

Tuesday: 4 - 5:30 pm & Friday: 2:30 - 4 pm

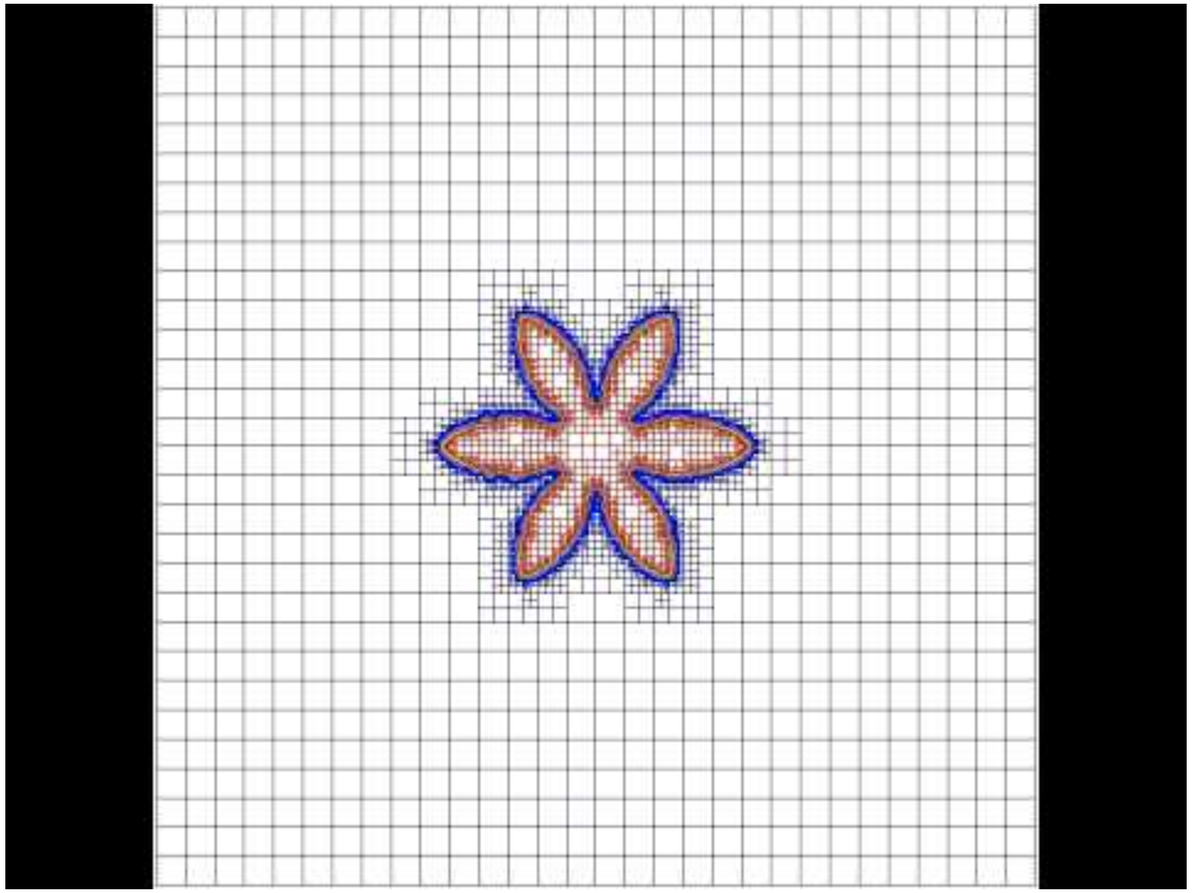
Anuj Goyal

Phenomenon: Water turning into ice



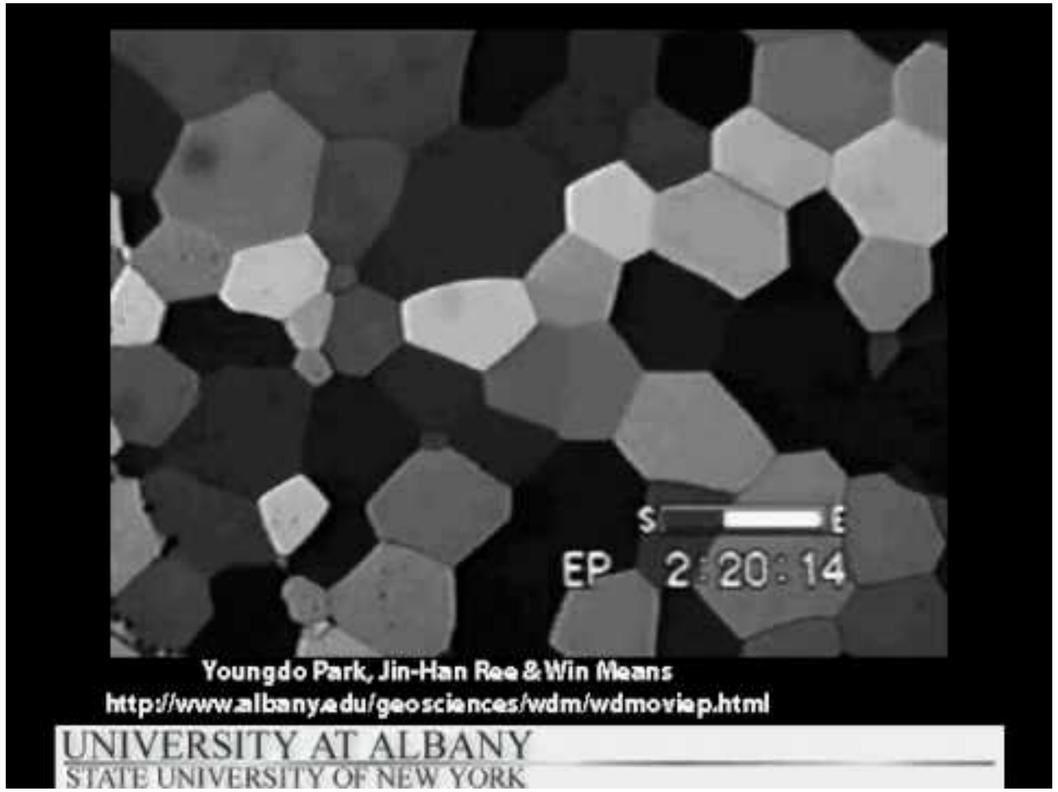
Method: Molecular Dynamics Simulation

Phenomenon: Snowflake formation



Method: 2D Phase Field Simulation

Geological Science: Grain Growth (Octachloropropane)



Method: Microstructure Modeling

Computational Model vs Simulation?

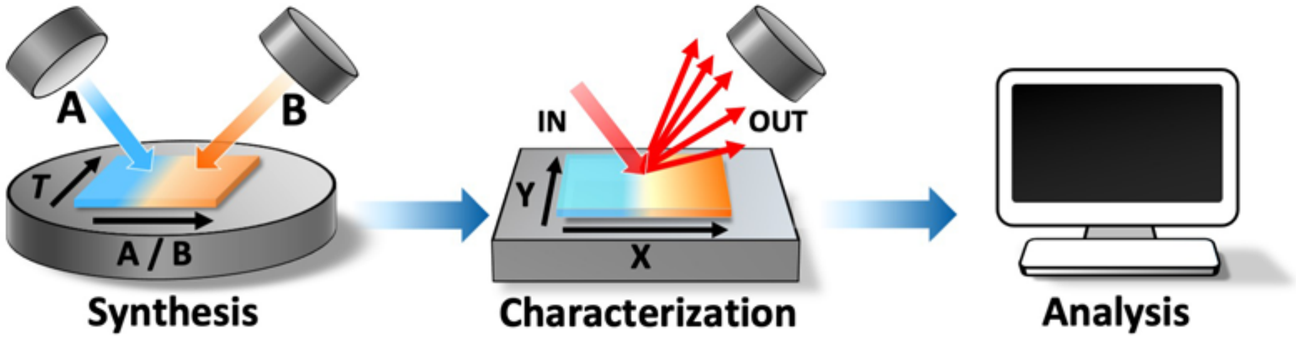
How do we build and characterize materials?

How do we build and characterize materials?

Experiments

- Traditional approach
- To optimize requires trial and error as well as time

How do we build and characterize materials?

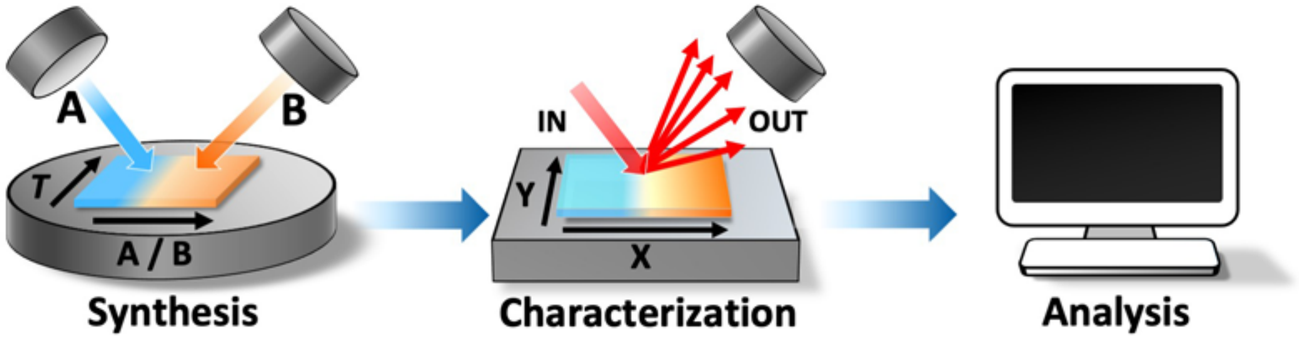


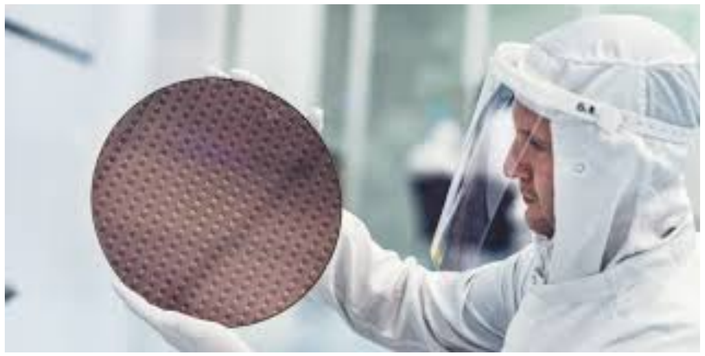
Experiments

- Traditional approach
- To optimize requires trial and error as well as time

Image reference: NREL

How do we build and characterize materials?



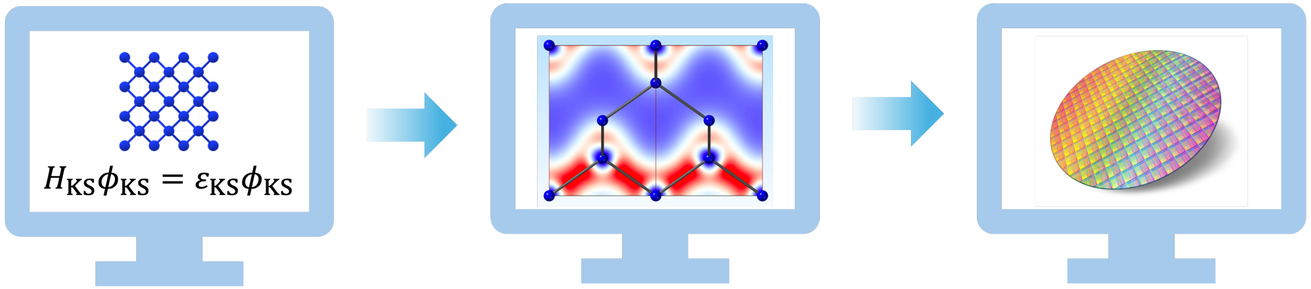


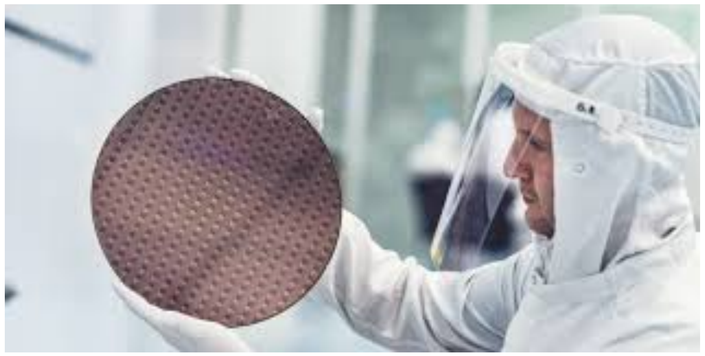
Materials Design & Discovery

Experiments

- Traditional approach
- To optimize requires trial and error as well as time

Why Computational Materials Modelling is Important?



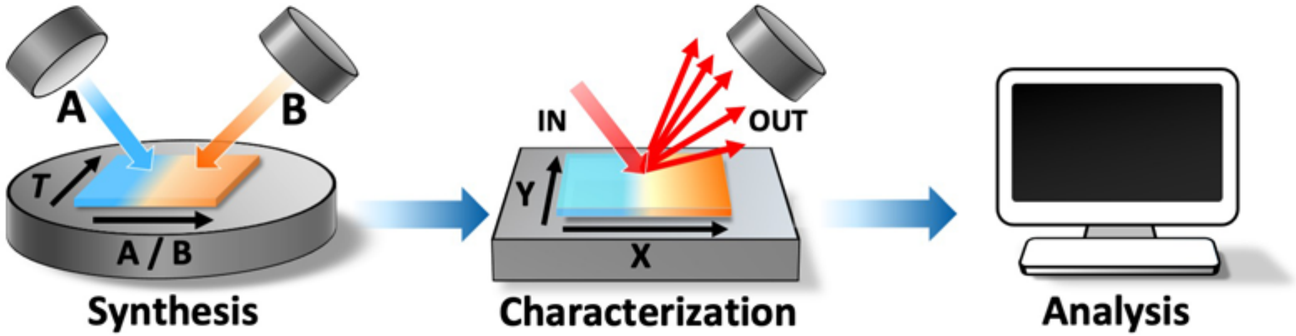


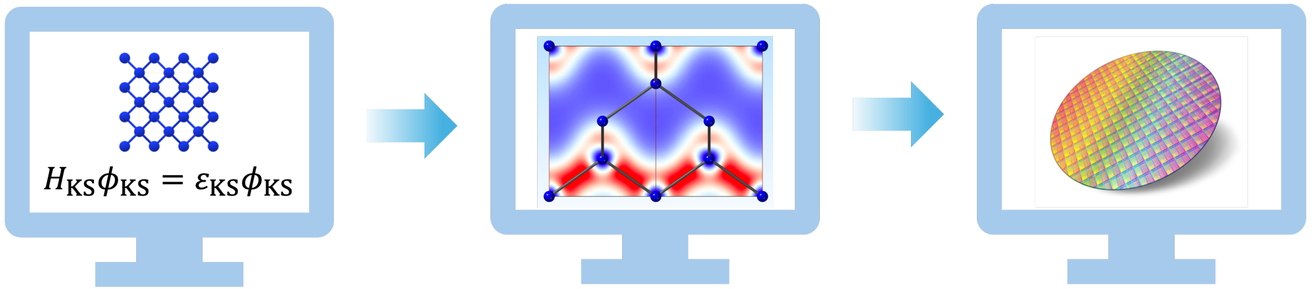
Materials Design & Discovery

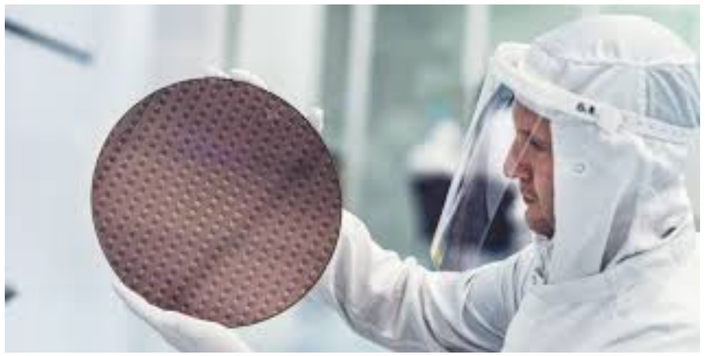
Computational Materials Modelling

- Complimentary to experiments
- Accelerated by availability of supercomputing resources

Why Computational Materials Modelling is Important?





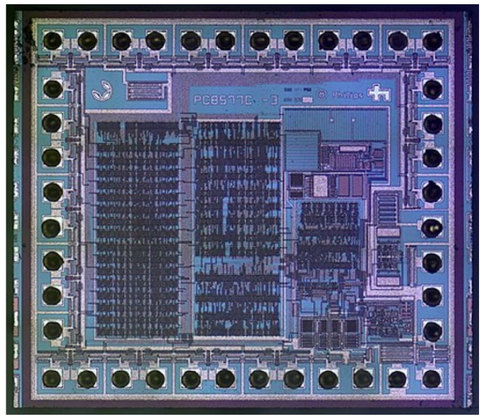


Materials Design & Discovery

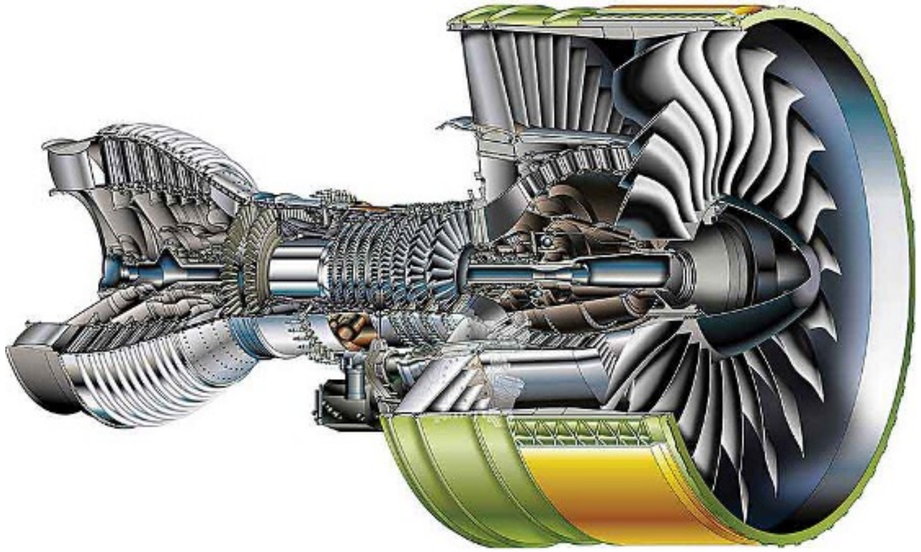
Computational materials modelling has emerged as a complementary approach to traditional materials discovery and design

Challenges in Materials Modelling

Integrated Circuit



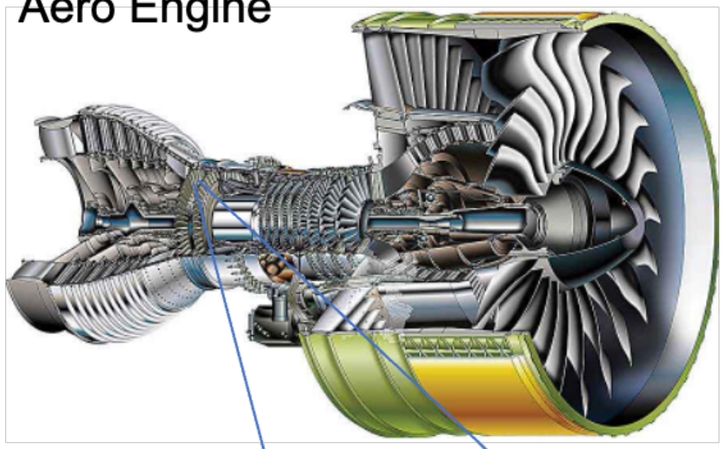
Aero Engine



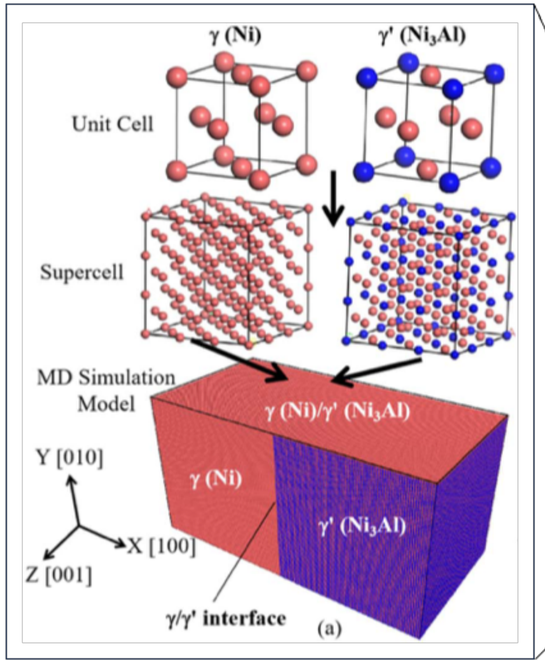
Too complex to model ….

Image reference: Wikipedia

Aero Engine

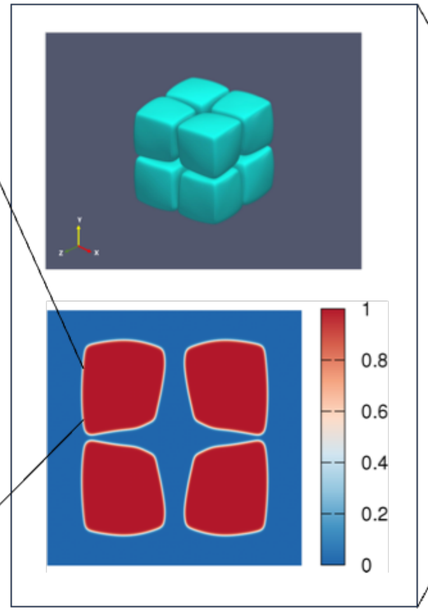


Atomic structure



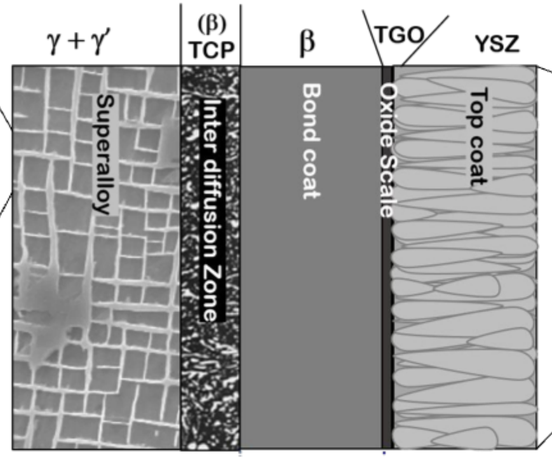
Molecular Dynamics (MD)

γ and γ’ phases



Phase field Simulations

Thermal Barrier Coating



Exp. Microstructure

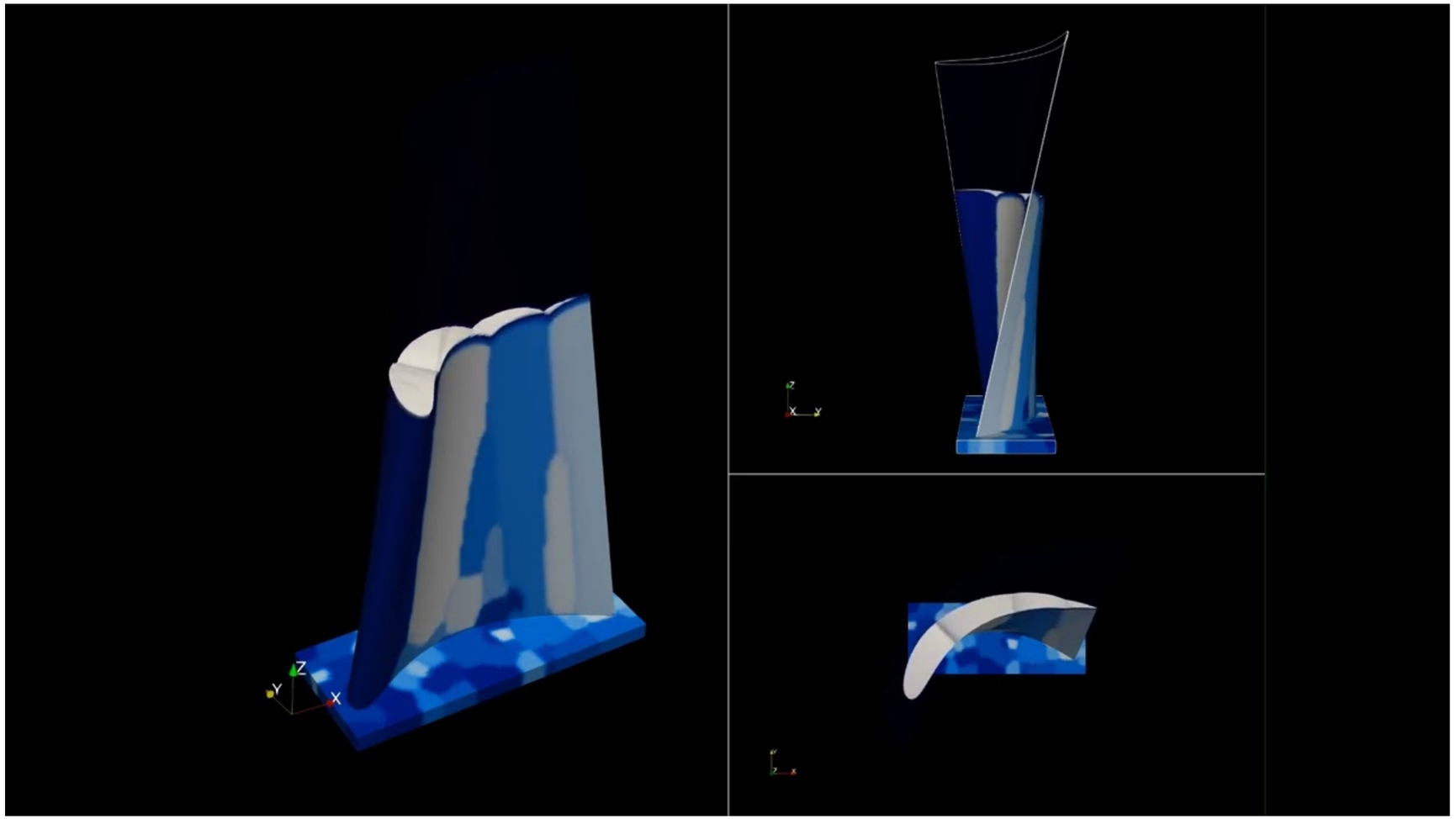


Turbine blade

~10 nm | ~1 μm | ~ 100 μm | ~ 10 cm

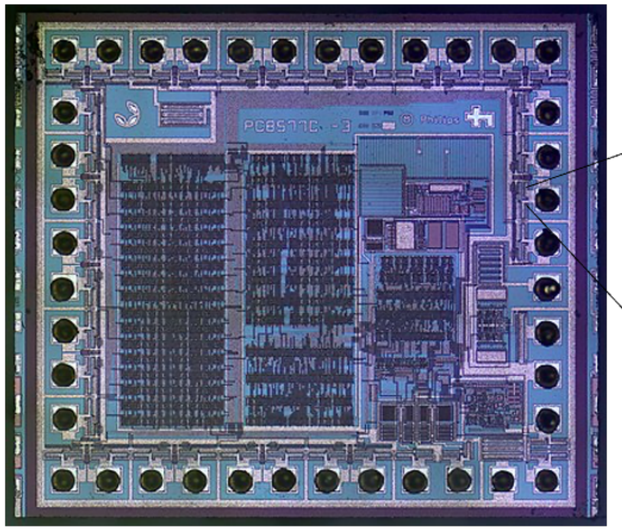
Image inspired from PhD Thesis (N. Esakkiraja, IISc, 2020)

Grain Growth in a Turbine Blade

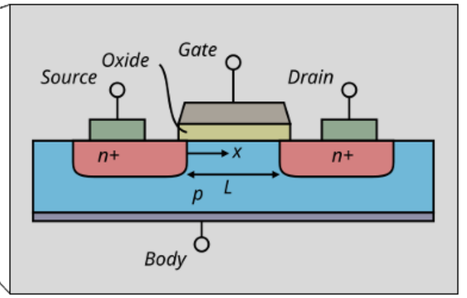


Method: Finite Element Method

Integrated Circuit

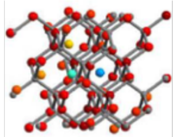
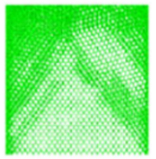
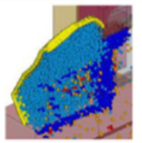
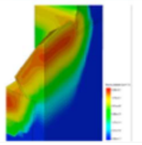


Transistor

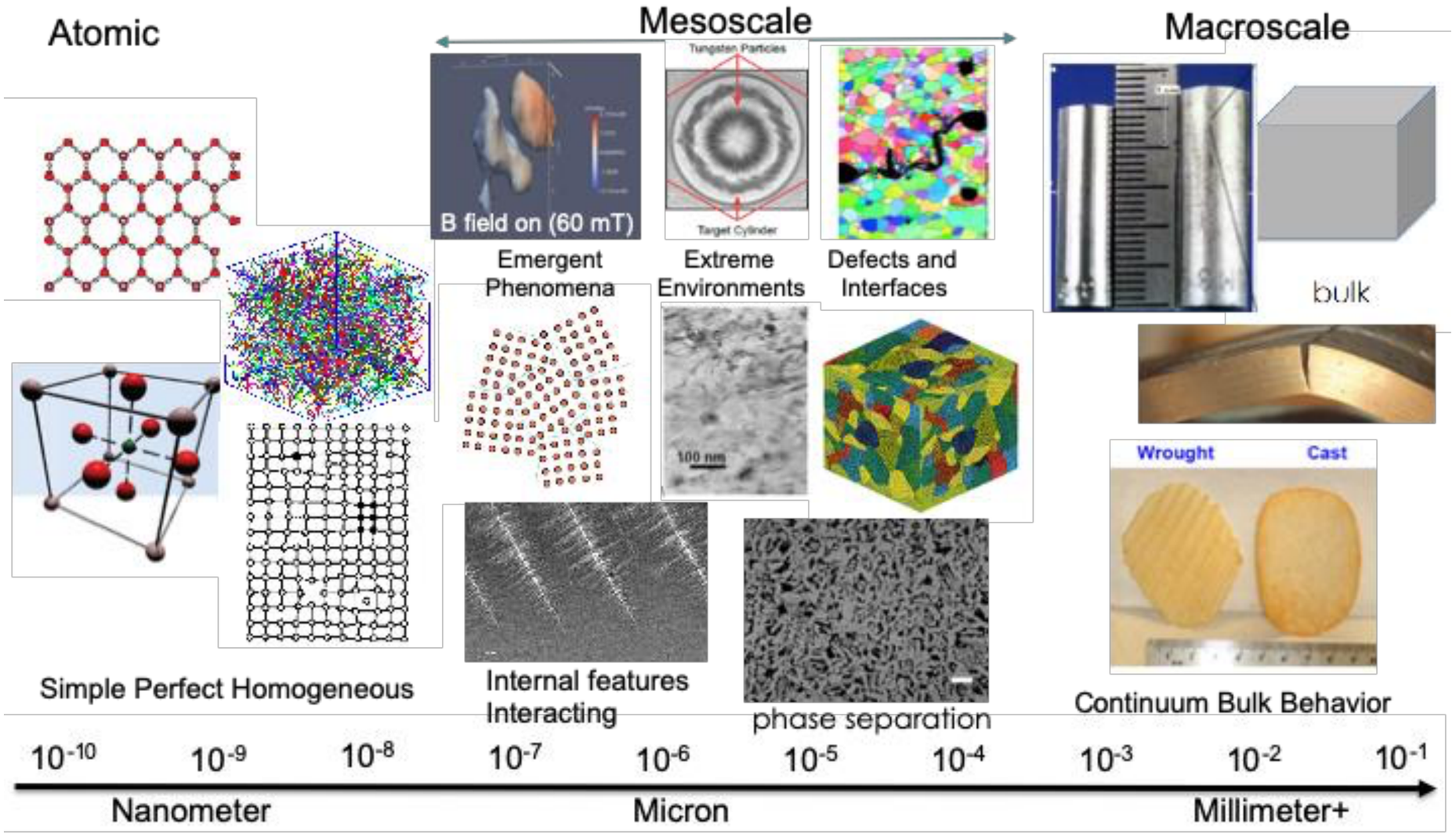


mm nm

Multiple length and time scale in Computational Modelling

Modeling Technique	Visualization	Structure Size	Times Simulated	CPU Time
Ab initio		~10^2 atoms	~10^{-2} ns	Slowest
Classical molecular dynamics		~10^5 atoms	~1 ns	
Lattice kinetic Monte Carlo		Part of device	Part of process	
Kinetic Monte Carlo		Complete device	Complete process	
Continuum		Complete device	Complete process	Fastest

Courtesy: N. Xographos et al. Materials Science in Semiconductor Processing (2017)

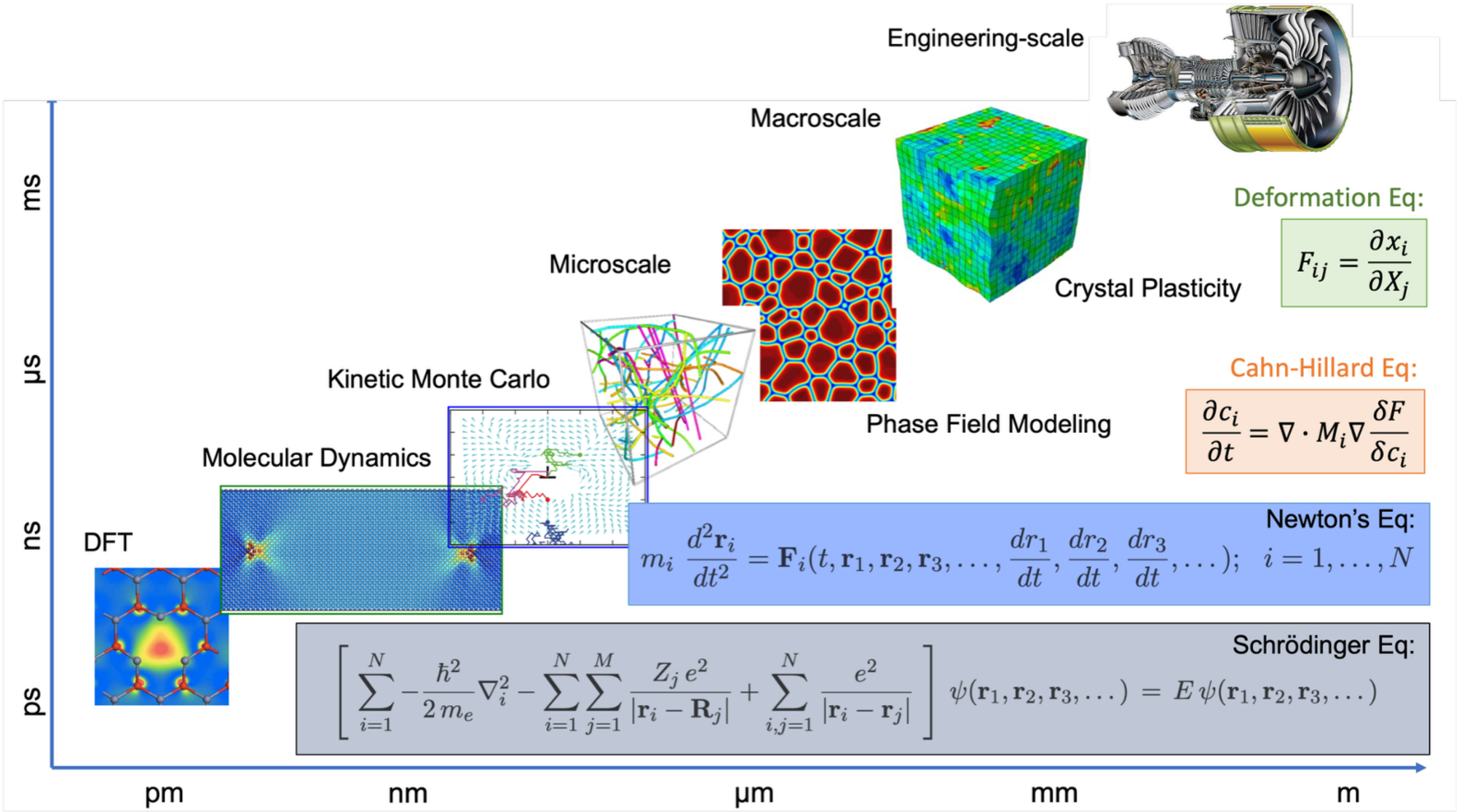


Courtesy: US Department of Energy (CC-BY)

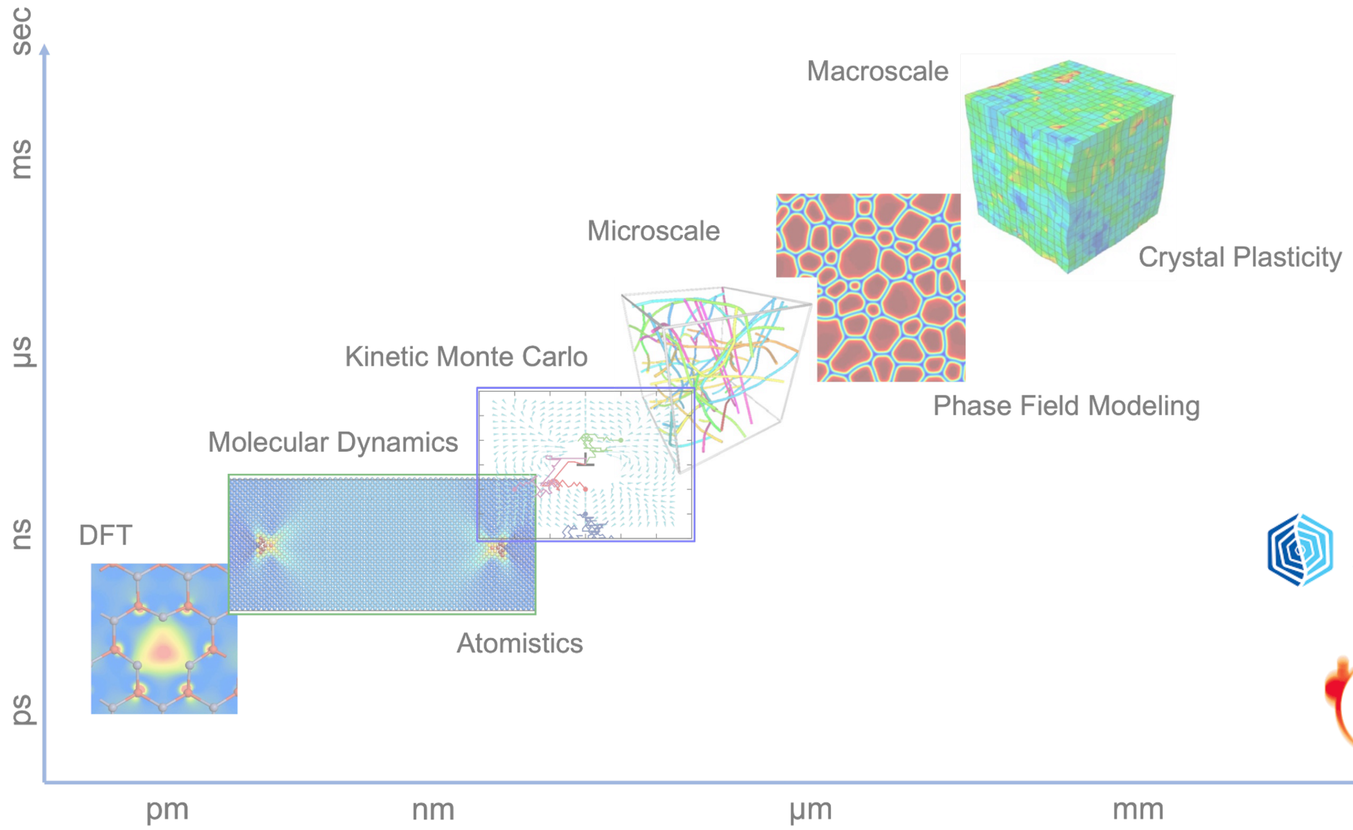
Physical Laws and Constitutive Differential Equations

Every phenomenon in nature can, in principle, be described using laws of physics which are usually formulated as partial differential equations

Physical Laws and Constitutive Differential Equations



Growing Computational Modelling Community ….



Syllabus

Foundation to Numerical Methods:

- Solving system of linear and non-linear equations
- Finding roots of non-linear equations
- Numerical Differentiation and Integration

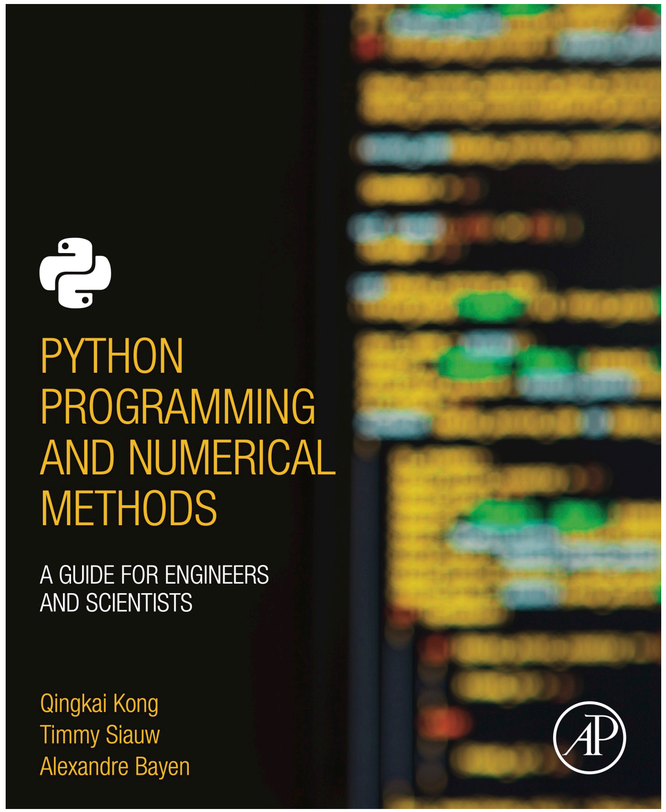
Numerical Methods:

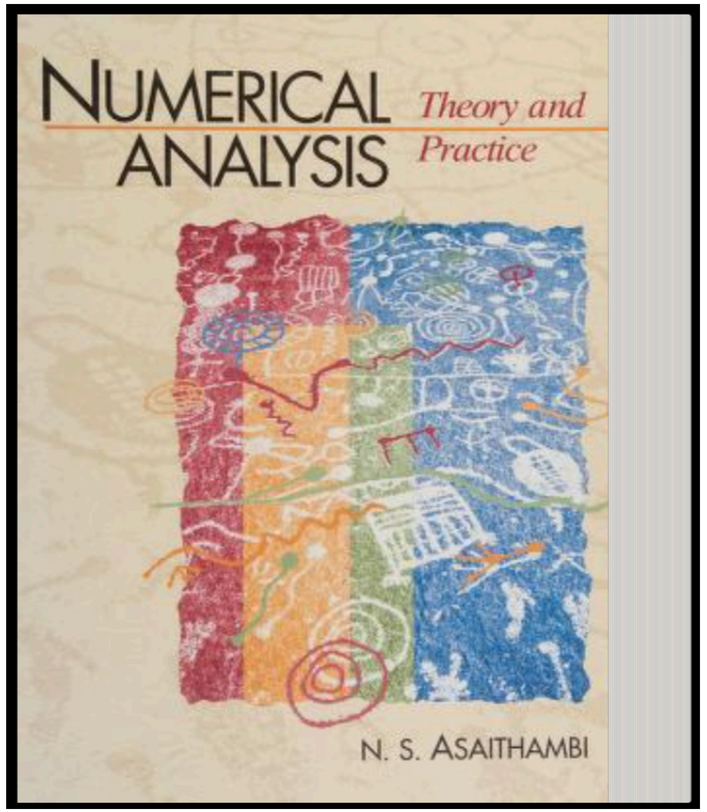
- Ordinary and Partial Differential Equations in Materials Engineering
- Initial and Boundary Value Problems
- Finite Difference Methods
- Fourier Method

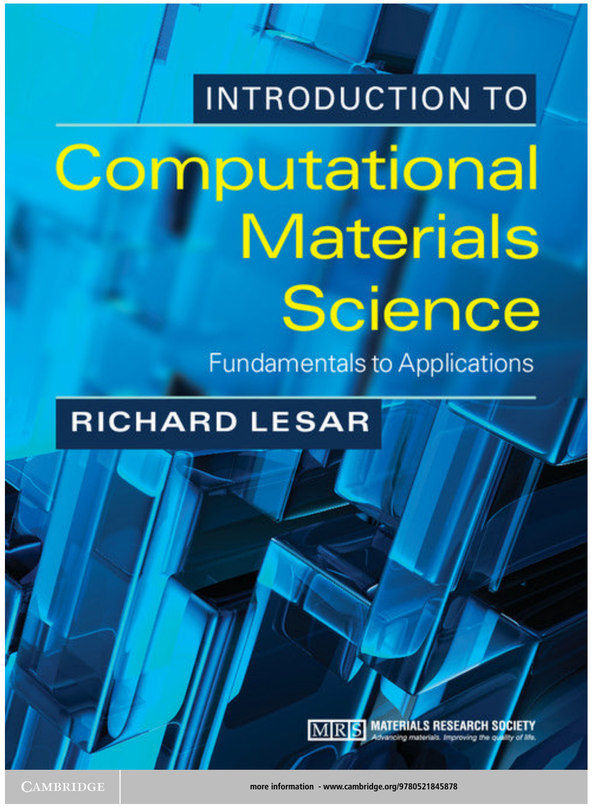
Application of Numerical Methods:

- Monte-Carlo Method
- Molecular Dynamics
- Ab initio Methods

Some of the References ....







Evaluation

Class presence & Discussion: 10%

Originality & Ethics: 15%

Assignments/Quizzes/Project: 75%

Office hours: Fridays (1:30 - 2:30 pm)